Crystals

Many minerals and other chemical compounds naturally form crystals. These glittering geometric structures are often created when molten or dissolved minerals slowly turn solid, giving time for their atoms to form organized structures. Some of the most beautiful crystals may be cut and polished into valuable gems.

CRYSTAL STRUCTURE

The atoms or molecules that form minerals cling to each other by electrical forces. Depending on the elements involved, they form different three-dimensional patterns. For example, common salt is made of chlorine and sodium atoms that bind together in a cubic structure. This defines the shape of salt crystals (right), which are also cubic.

ICE CRYSTALS

Frost is crystallized water, formed as liquid water cools to below its freezing point. The V-shaped, or triangular, water molecules latch onto each other as they cool. They form six-sided patterns that grow into six-sided, or hexagonal, crystals. Many mineral crystals form in the same way, but at higher temperatures. Quartz crystals start to form at about 3,090°F (1,700°C).

▼ WINTER FROST
The ice crystals glittering on this leaf have formed after an overnight frost.